

Ve 216: Introduction to Signals and Systems

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Based on Lecture Notes by Prof. Jeffrey A. Fessler and Prof. Yong Long

Outline

- 1 2. CT LTI Systems
 - Overview
 - Introduction
 - Techniques for the analysis of linear systems
 - Impulse response: mathematical and physical introduction
 - Impulse representation of CT signals (2.2.1)
 - Convolution for CT LTI systems (2.2.2)
 - Properties of convolution and LTI systems (2.3)
 - LTI system properties via impulse response (2.3.4-7)
 - Step response (2.3.8)
 - CT systems described by differential equation models (2.4)
 - Summary

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● T-1 Causal LTI systems (2.3.6)

● T-2 Memory (2.3.4)

● A-2 Stability of LTI systems (2.3.7)

● A-3 Invertibility of LTI systems (2.3.5)

● Step response (2.3.8)

● CT systems described by differential equation models (2.4)

● Solution of linear constant-coefficient diffeqs (2.4.1)

● Summary

Overview

- impulse response
- convolution (graphical, properties, LTI interconnection)
- differential equation (diffeq) systems (important class of LTI systems)
- Skip: 2.1, 2.4.2, 2.5.3 (the doublets part, but do read about unit ramp signal)

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Linear time-invariant (LTI) systems

Our primary focus hereafter will be on CT **linear time-invariant (LTI)** systems. This chapter is about analyzing such systems.

Why analysis? So far we only have **input-output relationships**. For a given system could compute $y(t)$ for a given $x(t)$, but it would be very difficult to design filters etc. by such trial-and-error.

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Linear and time-invariant system

- Recall, for a **linear** system:

$$a_1 x_1(t) + a_2 x_2(t) \xrightarrow{\mathcal{T}} y(t) = a_1 y_1(t) + a_2 y_2(t)$$

where $x_1(t) \xrightarrow{\mathcal{T}} y_1(t)$ and $x_2(t) \xrightarrow{\mathcal{T}} y_2(t)$.

- Recall, for a **time-invariant** system,

$$\text{if } x(t) \xrightarrow{\mathcal{T}} y(t), \quad \text{then } x(t - t_0) \xrightarrow{\mathcal{T}} y(t - t_0).$$

These two properties greatly simplify analysis of systems!

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LTI overview (1)

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$$x(t) \rightarrow \boxed{\text{LTI with impulse response } h(t)} \rightarrow y(t) = x(t) * h(t),$$

where $\delta(t) \xrightarrow{\mathcal{T}} h(t)$.

$$\boxed{\text{Input-output relationship: } y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau.}$$

Remarkably, the input-output relationship for any LTI system is given by the above **convolution integral**.

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Input-output relationship: $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$

- Conversely, any system whose input-output relationship can be expressed in the above form is an LTI system (easy to verify).
- An LTI system is completely described by its impulse response $h(t)$.
- We can determine the response $y(t)$ due to any input signal $x(t)$ by performing the convolution of the input signal with the impulse response.

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Techniques for the analysis of linear systems

General strategy:

- 1 **Decompose** input signal $x(t)$ into a **weighted sum of elementary functions** $x_k(t)$, *i.e.* $x(t) = \sum_k c_k x_k(t)$
Sometimes this decomposition is itself of interest in terms of studying the signal properties (*e.g.* Fourier analysis).
- 2 **Determine response** of system to each elementary function (this should be easy from input-output relationship):

$$x_k(t) \xrightarrow{\mathcal{T}} y_k(t)$$

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Elementary functions

Two particularly good choices for the elementary functions $x_k(t)$:

① **impulse functions** $\delta(t - \tau)$

② **complex exponentials** $e^{j\omega t}$ (later).

- **Linearity** leads to the superposition property, which simplifies the analysis.
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Impulse response

Definition

Impulse response $h(t)$ of an LTI system is defined as the response of the system to an input signal that is unit impulse.

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Impulse response: example

Example

Find the impulse response of the moving average system

$$y(t) = \frac{1}{T} \int_{t-T}^t x(\tau) d\tau. \text{ (Verify that it is an LTI system.)}$$

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$$h(t) = \frac{1}{T} \int_{t-T}^t \delta(\tau) d\tau = \begin{cases} 1/T, & t-T \leq 0 \leq t \\ 0, & \text{otherwise} \end{cases}$$

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A toy car example (1)

Example

A toy car on a carpet. The input signal $x(t)$ is the applied force, and the output signal $y(t)$ is the velocity of the car.

force $x(t)$ → car system → velocity $y(t)$

- If at time $t = 0$ someone hits the (previously stationary) car with a hard stick, then the car begins moving with some velocity.
- But friction gradually decreases the velocity towards zero. So the impulse response is something like:

$$h(t) = ae^{-t/b}u(t), \text{ (Picture).}$$

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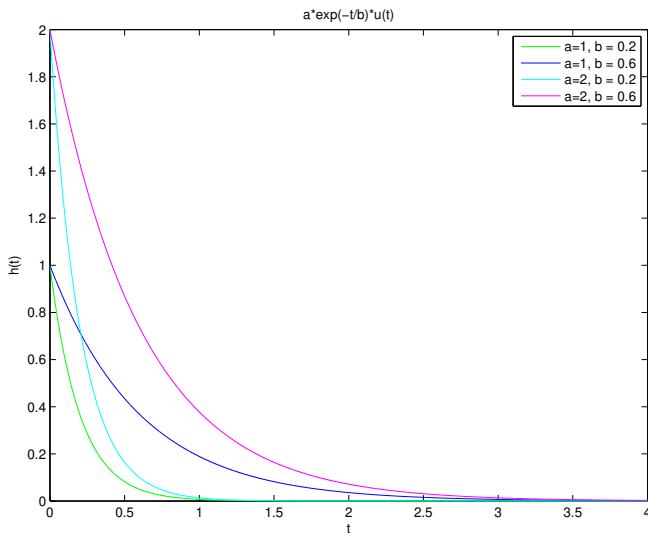
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A toy car example (3)

$$h(t) = ae^{-t/b}u(t), .$$

Question

- 1 *What is the constant a related to?*
- 2 *What is the constant b related to?*

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Question

- 1 What is the constant a related to? *a is the reciprocal of the mass of the car.*
- 2 What is the constant b related to? *Friction.*

A toy car example (4)

This toy car system is **time-invariant**. If we hit the car with the stick at time $t = t_0$ instead, then the velocity would look like

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The response will be $y(t) = h(t) + h(t - 3)$.

*This is the **linearity and time-invariant properties in action**.*

A toy car example (5)

The key question that is answered in this chapter is the following.

- Suppose that, instead of hitting the car (like an impulse), we give it a gentle push for a couple of seconds. What does the output signal (velocity vs time) look like?
- Now we are considering $x(t) = \text{rect}\left(\frac{t-1}{2}\right)$ for example, and we want to find $y(t)$.

We know the response of the system to any impulse. We want to find the response of the system to a rect signal (or a general input $x(t)$ for that matter). If only we could somehow express $x(t)$ as a “sum of delta functions”, then we could use the LTI properties to say that the output is the “sum of the responses” due to all those delta functions.

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Superposition of impulse function

Question

How to represent $x(t)$ as a “superposition” of impulse functions?

One way is to use the **sifting property** of the unit impulse function:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau.$$

$$\text{sifting property } \int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0).$$

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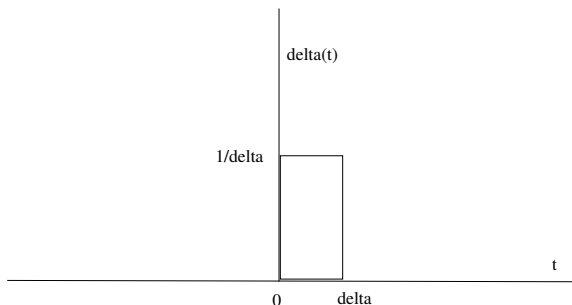
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Superposition of impulse function

The following representation may be more **intuitive**. Practical impulse function, defined for any $\Delta > 0$:

$$\delta_{\Delta}(t) \triangleq \begin{cases} 1/\Delta, & 0 < t < \Delta \\ 0, & \text{otherwise.} \end{cases} = \frac{1}{\Delta} \text{rect}\left(\frac{t}{\Delta} - \frac{1}{2}\right)$$



Impulse representation of CT signals (2)

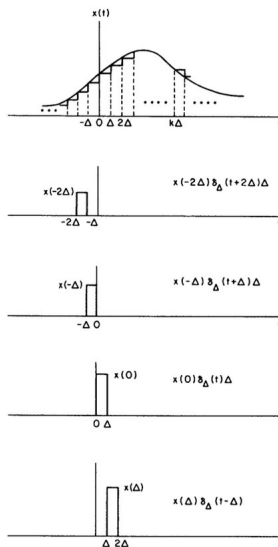
We can approximate any signal $x(t)$ by a **stairstep** signal formed from these pulse functions as follows

$$x(t) \approx \hat{x}(t) \triangleq \sum_{k=-\infty}^{\infty} x(k\Delta) \Delta \delta_{\Delta}(t - k\Delta).$$

Formally: $x(t) = \lim_{\Delta \rightarrow 0} \hat{x}(t)$.

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau.$$

Impulse representation of CT signals (3)



(MIT Lecture Notes 4-5)

Outline

1

2. CT LTI Systems

- Overview
- Introduction
- Techniques for the analysis of linear systems
- Impulse response: mathematical and physical introduction
- Impulse representation of CT signals (2.2.1)
- **Convolution for CT LTI systems (2.2.2)**
- Properties of convolution and LTI systems (2.3)
- LTI system properties via impulse response (2.3.4-7)
 - T-1 Causal LTI systems (2.3.6)
 - T-2 Memory (2.3.4)
 - A-2 Stability of LTI systems (2.3.7)
 - A-3 Invertibility of LTI systems (2.3.5)
- Step response (2.3.8)
- CT systems described by differential equation models (2.4)
 - Solution of linear constant-coefficient diffeqs (2.4.1)
- Summary

Response of practical impulse function

Question

Let $h_{\Delta}(t)$ denote the response of the system to the practical impulse function $\delta_{\Delta}(t)$, i.e.

$$\delta_{\Delta}(t) \xrightarrow{\mathcal{T}} h_{\Delta}(t).$$

- *What happens to δ_{Δ} as $\Delta \rightarrow 0$?*
- *What happens to h_{Δ} as $\Delta \rightarrow 0$?*

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- What happens to δ_{Δ} as $\Delta \rightarrow 0$? $\delta_{\Delta}(t) \rightarrow \delta(t)$.
- What happens to h_{Δ} as $\Delta \rightarrow 0$? $h(t) = \lim_{\Delta \rightarrow 0} h_{\Delta}(t)$.

Convolution for CT LTI systems (1)

If the system $\mathcal{T}$ is LTI, then **shifting** the input signal causes a shifted output signal

$$\delta_{\Delta}(t - k\Delta) \xrightarrow{\mathcal{T}} h_{\Delta}(t - k\Delta).$$

superposition property of a LTI system

$$\hat{x}(t) \triangleq \sum_{k=-\infty}^{\infty} x(k\Delta) \Delta \delta_{\Delta}(t - k\Delta) \xrightarrow{\mathcal{T}} \hat{y}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta) \Delta h_{\Delta}(t - k\Delta).$$

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Convolution for CT LTI systems (2)

On the input side, when we take the limit

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

On the output side, when we take the limit get:

$$\begin{aligned} y(t) &= \lim_{\Delta \rightarrow 0} \hat{y}(t) = \lim_{\Delta \rightarrow 0} \sum_{k=-\infty}^{\infty} x(k\Delta) \Delta h_{\Delta}(t - k\Delta) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \end{aligned}$$

convolution integral: $x(t) \xrightarrow{\mathcal{T}} y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau.$

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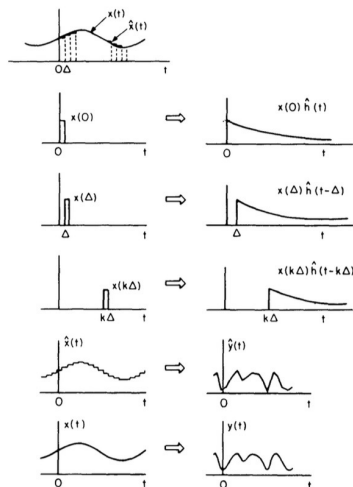
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Convolution for CT LTI systems (4)



(MIT, Lecture 4.7)

Convolution for CT LTI systems (5)

The response of an LTI system with impulse response $h(t)$ to an **arbitrary** input signal $x(t)$ is given by the convolution integral, which we also denote

$$y(t) = x(t) * h(t) = (x * h)(t).$$

This input-output relationship shows that **with the impulse response $h(t)$, you can compute the output $y(t)$ for any input signal $x(t)$.**

An LTI system is completely characterized by its impulse response $h(t)$.

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Convolution for CT LTI systems (6)

Let us now check that the term “impulse response” is appropriately named.

Example

Suppose the input signal is $\delta(t - 5)$. Find the output signal in terms of the impulse response $h(t)$.

By the sifting property of $\delta(t)$.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau = \int_{-\infty}^{\infty} \delta(\tau - 5)h(t-\tau) d\tau = h(t - 5)$$

Thus, the convolution integral confirms that the response of the system to a unit impulse at time $t = 5$ is the impulse response delayed to “start” at $t = 5$.

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Computing convolution (1)

Skill: *convolving*.

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

Recipe:

- 1 Fold: fold $h(\tau)$ about $\tau = 0$ to get $h(-\tau)$
- 2 Shift: shift $h(-\tau)$ by t to get $h(t - \tau)$
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Repeat for all possible t ; generally breaks in to a few intervals.

Mathematically, replace t with $t - \tau$ to complete step 1 and 2.

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Computing convolution (2)

Question

Can we fold and shift $x(t)$ instead of $h(t)$?

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Can we fold and shift $x(t)$ instead of $h(t)$?

We can fold and shift either $h(t)$ or $x(t)$ (commutative property of convolution).

*More often we use $h(t)$ since we will focus on **causal** systems, so folding and shifting $h(t)$ is particularly natural.*

Toy car example (1)

Example

Continue the toy car problem, but simplify by letting $b = 1$ so that $h(t) = e^{-at}u(t)$.

Find the response of the input $x(t) = u(t)$ (a steady push beginning at time 0).

Solution

$$\begin{aligned}
 y(t) &= x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \\
 &= \int_{-\infty}^{\infty} u(\tau) e^{-a(t-\tau)} u(t - \tau) d\tau \\
 &= u(t) \int_0^t e^{-a(t-\tau)} d\tau \quad (\tau \geq 0, t - \tau \geq 0 \implies 0 \leq \tau \leq t) \\
 &= e^{-at} \left(\frac{1}{a} e^{a\tau} \Big|_0^t \right) u(t) \\
 &= \frac{1}{a} (1 - e^{-at}) u(t)
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(**Picture**)(Example 2.6, p. 98) Video (MIT, Lecture 4, 32.33min)

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Toy car example (2)

Example

Now return to the problem determining response of the toy car to a short push: $x(t) = \text{rect}\left(\frac{t-1}{2}\right)$ and $h(t) = e^{-t}u(t)$.

Solution (1)

Solution

① **Fold:** $h(-\tau) = e^{-(-\tau)}u(-\tau) = e^{\tau}u(-\tau)$

② **Shift:** $h(t - \tau) = e^{-(t-\tau)}u(t - \tau)$

③ **Multiply:** $x(\tau)h(t - \tau) = \text{rect}\left(\frac{\tau-1}{2}\right) e^{-(t-\tau)}u(t - \tau)$

The product is nonzero only if $t \geq 0$, and even then the “shape” depends whether $t \in [0, 2]$ or $t > 2$.

Three cases to consider (we are making a braces function!).

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Solution (2)

- For $t < 0$, the integral $y(t) = 0$.
- For $0 < t < 2$ the integral is (using $\nu = -(t - \tau)$):

$$\int_0^t 1 e^{-(t-\tau)} d\tau = \int_{-t}^0 e^{\nu} d\nu = 1 - e^{-t}.$$

- For $t \geq 2$ the integral is

$$\int_0^2 1 e^{-(t-\tau)} d\tau = e^{-(t-2)}(1 - e^{-2}).$$

- Combining:

$$y(t) = \begin{cases} 0, & t \leq 0 \\ 1 - e^{-t}, & 0 < t < 2 \\ e^{-(t-2)}(1 - e^{-2}), & t \geq 2. \end{cases}$$

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Video (MIT, Lecture 4, 35min)

Toy car example (3)

Example

A toy car on a carpet. The input signal $x(t)$ is the applied force, and the output signal $y(t)$ is the velocity of the car.

force $x(t) \rightarrow$ car system $\rightarrow$ velocity $y(t)$

- A gentle push for a couple of seconds. $x(t) = \text{rect}\left(\frac{t-1}{2}\right)$.
- Impulse response $h(t) = e^{-t}u(t)$.
- Output velocity:

$$y(t) = \begin{cases} 0, & t \leq 0 \\ 1 - e^{-t}, & 0 < t < 2 \\ e^{-(t-2)}(1 - e^{-2}), & t \geq 2. \end{cases}$$

Video (MIT, Lecture 4, 36.38min)

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Toy car example (4)

Question

- *What affects the rise portion?*
- *What affects the decay portion?*

Toy car example (4)

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- *What affects the rise portion?*
The input signal (how hard we push and for how long).
forced response
- *What affects the decay portion?*

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- *What affects the rise portion?*
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- *What affects the decay portion?*
The system properties (decay due to friction losses). **natural response**

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Commutative property (2.3.1)

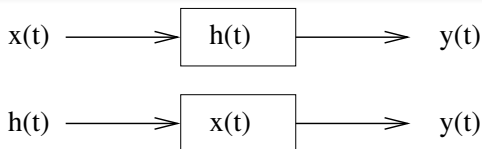
Skill: Use properties to simplify LTI systems.

Property

commutative property

$$x(t) * h(t) = h(t) * x(t).$$

Commutative



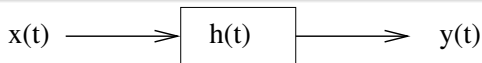
Commutative property (2.3.1)

Skill: Use properties to simplify LTI systems.

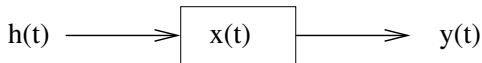
Property

commutative property

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Commutative



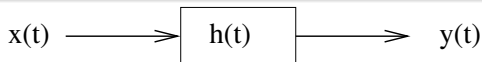
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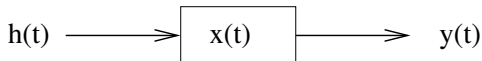
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commutative property

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Commutative



Commutative property: proof

Question

Show the commutative property?

Commutative property: proof

Question

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$$\begin{aligned}x(t) * h(t) &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \\&= \int_{-\infty}^{\infty} x(t - s) h(s) ds \quad (s = t - \tau \rightarrow \tau = t - s) \\&= h(t) * x(t)\end{aligned}$$

Associative property (2.3.3)

Property

Associative property

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

This property holds in general for any number of systems connected in [series](#). So the following notation is acceptable:

$$h(t) = h_1(t) * h_2(t) * \cdots * h_k(t).$$

In particular:

$$\begin{aligned}(x * h_1) * h_2 &= x * (h_1 * h_2) \\ &= x * (h_2 * h_1) \\ &= (x * h_2) * h_1\end{aligned}$$

Associative property (2.3.3)

Property

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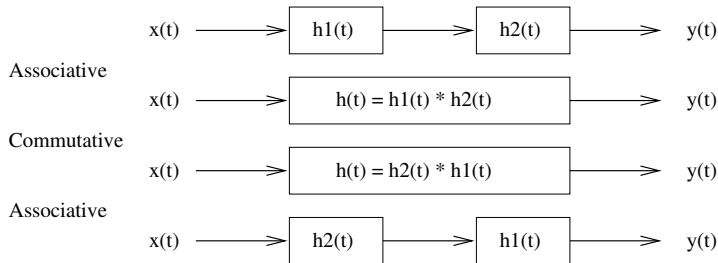
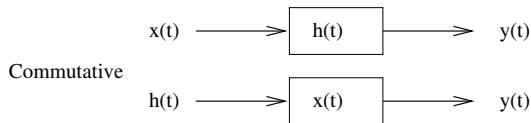
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Commutative and associative property

The order of serial connection of LTI systems does not affect the overall impulse response.



Associative property: proof (1)

Question

Show the Associative property.

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

Associative property: proof (1)

Question

Show the Associative property.

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

let $y_1(t) = [x(t) * h_1(t)] * h_2(t)$ and $y_2(t) = x(t) * [h_1(t) * h_2(t)]$.
We must show $y_1(t) = y_2(t)$.

Associative property: proof (2)

$$\begin{aligned}y_1(t) &= \int_{-\infty}^{\infty} [x * h_1](\tau) h_2(t - \tau) d\tau \\&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(s) h_1(\tau - s) ds \right] h_2(t - \tau) d\tau \\&= \int_{-\infty}^{\infty} x(s) \left[\int_{-\infty}^{\infty} h_1(\tau - s) h_2(t - \tau) d\tau \right] ds \\&= \int_{-\infty}^{\infty} x(s) \left[\int_{-\infty}^{\infty} h_1(t - u - s) h_2(u) du \right] ds \quad (t - \tau = u \rightarrow \tau = t - u) \\&= \int_{-\infty}^{\infty} x(s) [h_1 * h_2](t - s) ds \\&= (x * [h_1 * h_2])(t) = y_2(t)\end{aligned}$$

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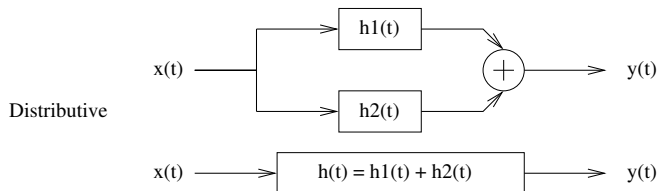
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Distributive property

Property

Distributive property

$$x(t) * [h_1(t) + h_2(t)] = [x(t) * h_1(t)] + [x(t) * h_2(t)]$$



Question

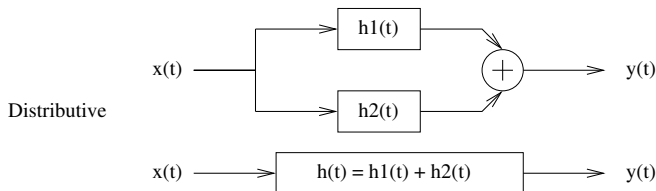
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Question

Show the distributive property.

Distributive property: proof

$$\begin{aligned}x(t) * [h_1(t) + h_2(t)] &= \int_{-\infty}^{\infty} x(t - \tau)[h_1(\tau) + h_2(\tau)] d\tau \\&= \int_{-\infty}^{\infty} x(t - \tau)h_1(\tau) d\tau + \int_{-\infty}^{\infty} x(t - \tau)h_2(\tau) d\tau \\&= x(t) * h_1(t) + x(t) * h_2(t)\end{aligned}$$

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Example

Example

Given $u(t) * h(t) = (1 - e^{-t})u(t)$, find an easier approach to previous problem of finding $y(t) = x(t) * h(t)$ where $x(t) = \text{rect}(\frac{t-1}{2})$ and $h(t) = e^{-t}u(t)$.

- 1 $x(t) = \text{rect}(\frac{t-1}{2}) = u(t) - u(t-2)$
- 2 $u(t) - u(t-2) \xrightarrow{\mathcal{T}} z(t) - z(t-2)$ where
 $u(t) \xrightarrow{\mathcal{T}} z(t) = h(t) * u(t) = (1 - e^{-t})u(t)$
- 3 Thus
 $y(t) = z(t) - z(t-2) = (1 - e^{-t})u(t) - (1 - e^{-(t-2)})u(t-2).$

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$$1 \quad x(t) = \text{rect}(\frac{t-1}{2}) = u(t) - u(t-2)$$

$$2 \quad u(t) - u(t-2) \xrightarrow{\mathcal{T}} z(t) - z(t-2) \text{ where} \\ u(t) \xrightarrow{\mathcal{T}} z(t) = h(t) * u(t) = (1 - e^{-t})u(t)$$

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$$\textcircled{1} \quad x(t) = \text{rect}\left(\frac{t-1}{2}\right) = u(t) - u(t-2)$$

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Properties of convolution and impulse functions (1)

Property

$$x(t) * \delta(t) = x(t)$$

impulse representation of a CT signal

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

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Properties of convolution and impulse functions (2)

Property

Delay property: $x(t) * \delta(t - t_0) = x(t - t_0)$

Question

Prove the above property.

Properties of convolution and impulse functions (2)

Property

Delay property: $x(t) * \delta(t - t_0) = x(t - t_0)$

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Solution

Solution

Proof.

$$\begin{aligned}x(t) * \delta(t - t_0) &= \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau - t_0) d\tau \\&= \int_{-\infty}^{\infty} x(\tau) \delta(t - t_0 - \tau) d\tau \\&= \int_{-\infty}^{\infty} x(\tau) \delta(t' - \tau) d\tau \quad (t' = t - t_0) \\&= x(t') = x(t - t_0)\end{aligned}$$

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Properties of convolution and impulse functions (3)

Property

$$\delta(t - t_0) * \delta(t - t_1) = \delta(t - t_0 - t_1)$$

Property

*If $y(t) = x(t) * h(t)$, then $x(t - t_0) * h(t - t_1) = y(t - t_0 - t_1)$.
(Due to time invariance of system.)*

Question

Prove the above property.

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Prove the above property.

Proof

$$\begin{aligned}x(t - t_0) * h(t - t_1) &= \int_{-\infty}^{\infty} x(\tau - t_0) h(t - \tau - t_1) d\tau \\&= \int_{-\infty}^{\infty} x(\tau') h(t - t_1 - (\tau' + t_0)) d\tau' \quad (\tau' = \tau - t_0) \\&= \int_{-\infty}^{\infty} x(\tau') h(t' - \tau') d\tau' \quad (t' = t - t_0 - t_1) \\&= y(t') = y(t - t_0 - t_1)\end{aligned}$$

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2. CT LTI Systems

- Overview
- Introduction
- Techniques for the analysis of linear systems
- Impulse response: mathematical and physical introduction
- Impulse representation of CT signals (2.2.1)
- Convolution for CT LTI systems (2.2.2)
- Properties of convolution and LTI systems (2.3)
- **LTI system properties via impulse response (2.3.4-7)**
 - T-1 Causal LTI systems (2.3.6)
 - T-2 Memory (2.3.4)
 - A-2 Stability of LTI systems (2.3.7)
 - A-3 Invertibility of LTI systems (2.3.5)
- Step response (2.3.8)
- CT systems described by differential equation models (2.4)
 - Solution of linear constant-coefficient diffeqs (2.4.1)
- Summary

LTI system properties via impulse response

Since an LTI system is completely characterized by its impulse response, we should be able to express the remaining four properties in terms of $h(t)$.

- T-1 causality
- T-2 memory
- A-2 stability
- A-3 invertibility

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T-1 Causal LTI systems (1)

Recall a system is **causal** iff output $y(t)$ depends only on present and past values of input.

For an LTI system with impulse response $h(t)$:

$$\begin{aligned}y(t) &= \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau \\&= \int_0^{\infty} h(\tau)x(t-\tau) d\tau + \int_{-\infty}^{0^-} h(\tau)x(t-\tau) d\tau.\end{aligned}$$

- ① The first term depends on present and past input values $x(t), x(t-\tau)$ for $\tau \geq 0$.
- ② The second term depends on future input values $x(t-\tau)$ for $\tau < 0$.

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T-1 Causal LTI systems (2)

Thus the system is causal iff the impulse response terms corresponding to the **second term** are zero. These terms are $h(\tau)$ for $\tau < 0$.

Definition

An LTI system is **causal** iff its impulse response $h(t) = 0$ for all $t < 0$.

In the causal case the convolution integral **simplifies** slightly since we can drop the right term above:

$$y(t) = \int_0^{\infty} h(\tau)x(t-\tau) d\tau = \int_{-\infty}^t x(\tau')h(t-\tau') d\tau' \text{ (using } \tau' = t - \tau \text{)}.$$

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Is the LTI system with $h(t) = u(t - t_0 - 5)$ causal?

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Solution

It is *causal* only if $h(t) = 0$ for all $t < 0$. $\Rightarrow t_0 + 5 \geq 0$.

T-1 Causal LTI systems (4)

Definition

A **causal signal** is a signal $x(t)$ which is zero for all $t < 0$.

If the input to a causal LTI system is a causal signal, then the output is simply

$$y(t) = \begin{cases} 0, & t < 0 \\ \int_0^t h(\tau)x(t-\tau) d\tau = \int_0^t x(\tau)h(t-\tau) d\tau, & t \geq 0. \end{cases}$$

MATLAB's `conv` function computes a discrete-time version of the above integral, for finite-duration $x(t)$ and $h(t)$.

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T-2 Memory (1)

Recall a system is **static** or **memoryless** if the output $y(t)$ depends only on the current input $x(t)$, not on previous or future values of the input signal.

For an LTI system with impulse response $h(t)$:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau,$$

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An LTI system is **memoryless** iff its impulse response is $h(t) = a\delta(t)$. Otherwise the system is **dynamic** (has memory).

In this case the response is

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There are two classes of dynamic systems.

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$$h(t) = \text{rect}(t)$$

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A-2 Stability of LTI systems (1)

Recall $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau$. Suppose $x(t)$ is a **bounded input signal**, i.e. $|x(t)| \leq M_x < \infty \forall t$. Under what conditions on $h(t)$ is the response $y(t)$ a bounded signal?

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau \right| \\ &\leq \int_{-\infty}^{\infty} |h(\tau)x(t-\tau)| d\tau \quad (\text{triangle inequality}) \\ &= \int_{-\infty}^{\infty} |h(\tau)||x(t-\tau)| d\tau \leq M_x \int_{-\infty}^{\infty} |h(\tau)| d\tau. \end{aligned}$$

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Necessary condition: find a single example of a bounded input $x(t)$ that causes an unbounded output $y(t) = x(t) * h(t)$ when $h(t)$ is not absolutely integrable.

$$y(0) = \int_{-\infty}^{\infty} h(\tau)x(0 - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau)x(-\tau) d\tau$$

Consider now the case of a bounded input signal specified as

$$x(t) = \operatorname{sgn}(h(-t)) = \pm 1$$

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A-2 Stability of LTI systems (3)

Definition

An LTI system is **BIBO stable** iff its impulse response is **absolutely integrable**, *i.e.* $\int_{-\infty}^{\infty} |h(t)| dt < \infty$.

Example

Integrator: $y(t) = \int_{-\infty}^t x(\tau) d\tau$.

What is impulse response? Is it stable?

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What is impulse response? Is it stable?

Solution (1)

Solution

Approach 1:

let $x(t) = \delta(t)$, then perform the running integral, i.e.,

$$h(t) = \int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

Solution (2)

Solution

Approach 2:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^{\infty} u(t - \tau)x(\tau) d\tau.$$

*Since this has the exact form of a convolution integral with $u(t - \tau)$ in the place where $h(t - \tau)$ goes, we can conclude that $h(t - \tau) = u(t - \tau)$, or equivalently simply that $h(t) = u(t)$.
This is a generally useful trick for finding $h(t)$!*

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Solution (3)

Solution

$$\int_{-\infty}^{\infty} |u(t)| dt = \int_0^{\infty} 1 dt = \infty, \text{ so } \textit{unstable}.$$

A-2 Stability of LTI systems (3)

Example

Is the moving average $h(t) = \frac{1}{T} \text{rect}(t/T - \frac{1}{2})$ stable?

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Is the moving average $h(t) = \frac{1}{T} \text{rect}(t/T - \frac{1}{2})$ stable?

$$\int_{-\infty}^{\infty} |h(t)| dt = \frac{1}{T} \int_0^T 1 dt = 1 < \infty. \text{ Thus } \textit{stable}.$$

A-2 Stability of LTI systems (4)

Example

Show the stability of the decaying sinusoid system with impulse response of $h(t) = e^{-t}(\cos t)u(t)$.

Solution

We can avoid a messy integral by thinking:

$$\begin{aligned}\int_{-\infty}^{\infty} |h(t)| dt &= \int_{-\infty}^{\infty} |e^{-t}(\cos t)u(t)| dt \\ &= \int_0^{\infty} e^{-t} |\cos t| dt \\ &\leq \int_0^{\infty} e^{-t} 1 dt \\ &= 1.\end{aligned}$$

Outline

1

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A-3 Invertibility of LTI systems (1)

Recall if a system $\mathcal{T}$ is **invertible**, then there exists a system $\mathcal{T}^{-1}$ such that

$$x(t) \rightarrow \boxed{\mathcal{T}} \rightarrow y(t) \rightarrow \boxed{\mathcal{T}^{-1}} \rightarrow z(t) = x(t).$$

Fact: if a system is LTI, then if it is also invertible then the inverse system is also LTI (Problem 2.50, textbook). So

$$x(t) \rightarrow \boxed{\text{LTI } h(t)} \rightarrow y(t) \rightarrow \boxed{\text{LTI } h_i(t)} \rightarrow z(t) = x(t),$$

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- The cascade of two LTI systems is also LTI, so can be characterized by its impulse response.
- If the input is $x(t) = \delta(t)$, then the output is

$$x(t) * h(t) * h_i(t) = \delta(t) * h(t) * h_i(t)$$

- Thus, if the system is invertible, then

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Definition

An LTI system is **invertible** iff there exist an inverse system whose impulse response $h_i(t)$ satisfies the following relationship with $h(t)$

$$h(t) * h_i(t) = \delta(t).$$

Example

Consider the delay system with $h(t) = 2\delta(t - 5)$. Is it invertible? If yes, find the impulse response of its inverse system.

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It is *invertible*. The inverse system has impulse response $h_i(t) = \frac{1}{2}\delta(t + 5)$.

Later, using *Fourier transforms*, we will see how to find $h_i(t)$ more generally, and how to determine when a given $h(t)$ does and does not correspond to an invertible system.

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The impulse response of an invertible LTI system is $h(t) = \delta(t) - e^{-t}u(t)$. Show a system with the impulse response of $h_i(t) = \delta(t) + u(t)$ is its inverse system.

Solution (1)

It suffices to show that $h(t) * h_i(t) = \delta(t)$.

$$\begin{aligned}
 & h(t) * h_i(t) \\
 = & [\delta(t) - e^{-t}u(t)] * [\delta(t) + u(t)] \quad (\text{distributive property}) \\
 = & \delta(t) * \delta(t) + \delta(t) * u(t) - (e^{-t}u(t) * \delta(t)) - (e^{-t}u(t) * u(t)) \\
 = & \delta(t) + u(t) - e^{-t}u(t) - \int_{-\infty}^{\infty} e^{-(t-\tau)} u(t-\tau) u(\tau) d\tau \\
 = & \delta(t) + (1 - e^{-t}) u(t) - \left(\int_0^t 1 e^{-(t-\tau)} d\tau \right) u(t) \\
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 = & \delta(t) + (1 - e^{-t}) u(t) - (1 - e^{-t}) u(t) \\
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Impulse response of an LTI system

Question

How can we find the impulse response $h(t)$ of an LTI system in practice?

- 1 One way is to put a “practical impulse” through the system and observe output. This approach can be difficult since a large input spike could drive some systems into nonlinear regime.
- 2 An alternative is to first find the **step response**, and then differentiate.

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Step response and impulse response

Question

Find the **step response**, and then differentiate to find the **impulse response** $h(t)$. Why does this work?

$$u(t) \rightarrow \boxed{d/dt} \rightarrow \delta(t) \rightarrow \boxed{\text{LTI}} \rightarrow h(t)$$

by associative/commutative property:

$$u(t) \rightarrow \boxed{\text{LTI}} \rightarrow s(t) \rightarrow \boxed{d/dt} \rightarrow h(t)$$

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Step response

Definition

The **(unit) step response** of an LTI system is the running integral of its impulse response, or

$$s(t) = \int_{-\infty}^{\infty} u(t - \tau)h(\tau) d\tau = \int_{-\infty}^t h(\tau) d\tau$$

$\delta(t) \xrightarrow{\mathcal{T}} h(t)$ called **impulse response**
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Step response

If the system is **causal**, then $h(t) = 0$ for $t < 0$ so

$$s(t) = \int_{-\infty}^t h(\tau) d\tau = \left[\int_0^t h(\tau) d\tau \right] u(t)$$

Relations:

$$\begin{aligned} s(t) &= \left[\int_0^t h(\tau) d\tau \right] u(t) \\ h(t) &= \frac{d}{dt} s(t) \end{aligned}$$

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Example

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Earlier for car problem we showed that step response was $s(t) = (1 - e^{-t})u(t)$. Find the impulse response.

Solution

Solution

The impulse response is

$$\begin{aligned}h(t) &= \frac{d}{dt}s(t) = \frac{d}{dt}[(1 - e^{-t})u(t)] \\&= \left[\frac{d}{dt}(1 - e^{-t})\right]u(t) + (1 - e^{-t})\left[\frac{d}{dt}u(t)\right] \\&= e^{-t}u(t) + (1 - e^{-t})\delta(t) \\&= e^{-t}u(t) + (1 - e^{-0})\delta(t) \quad \text{sampling property of } \delta(t) \\&= e^{-t}u(t),\end{aligned}$$

which indeed is the impulse response we had started with originally in that example.

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CT systems described by differential equation models (1)

- The **convolution integral** $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau$ looks fine on paper, but what about **hardware** (e.g. circuit) implementation?
- We might use **mathematical analysis** to design the “perfect” **impulse response** $h(t)$, but if there is no **physical system** that has that impulse response, then our design is of limited use.

Let us focus on **RLC circuits** for the moment, and see what types of input-output relationships (and hence what types of impulse response functions) can be realized.

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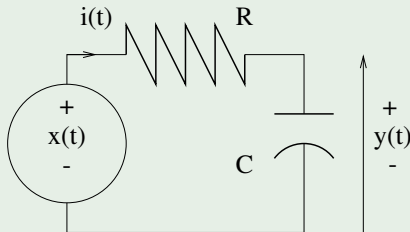
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CT systems described by differential equation models (2)

Example

The input signal $x(t)$ is the voltage source, and the output signal $y(t)$ is the voltage across the capacitor.



The current is $i(t) = (x(t) - y(t))/R$, and the current is also C times the derivative of the voltage across the capacitor:

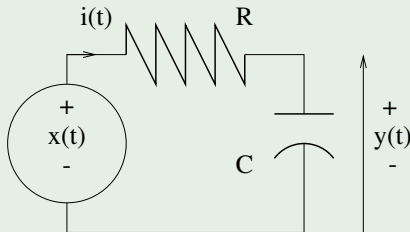
$i(t) = C \frac{d}{dt} y(t)$. Thus

$$\frac{x(t) - y(t)}{R} = C \frac{d}{dt} y(t) \quad \text{or} \quad y(t) + RC \frac{d}{dt} y(t) = x(t)$$

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Had we used a larger number of circuit elements, we would have found a differential equation (diffee) with more terms.

Definition

The input-output relationship for an RLC circuit (with a finite number of components) is a diffee of the following form, called a **linear constant-coefficient differential equation**:

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t)$$

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$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t).$$

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- We assume hereafter that that a_k 's and b_k 's are **real**.
- N is called the **order** of the system

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What is order of above RC circuit?

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Example

What is order of above RC circuit?

*It is called **first order**, since $N = 1$.*

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$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t).$$

In general there is **not a simple time-domain method** for finding the impulse response of such systems. We will do it systematically later using **frequency response** and the **partial fraction expansion (PFE)** method.

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1

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Solution of linear constant-coefficient diffeqs (1)

The general solution to a diffeq can be decomposed into the sum of two parts:

$$y(t) = y_h(t) + y_p(t).$$

- $y_h(t)$ is called the **homogeneous solution** or **zero-input solution** in the diffeq world, or the **natural response** in the systems world.
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Method of undetermined coefficients (1)

The **natural response** is found by solving the **zero-input** equation:

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = 0. \quad (1) \quad \boxed{e, z}$$

It was discovered long ago that the solutions to the above equation have components of the form

$$y(t) = Ce^{st}$$

for values $C \neq 0$ and s to be determined. Plugging into (1) yields

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Clearly this is only zero for all t if s satisfies

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i.e. if s is a root of the above **characteristic equation** or **characteristic polynomial**.

The characteristic equation has N roots, let us label them $s_1, \dots, s_N$. (They may be distinct or not, real or complex.)

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For the general case of an r -th order root s_i in the characteristic equation, the (almost) general natural response is

$$y_h(t) = \left[\sum_{j=1}^r C_j t^{j-1} \right] e^{s_i t} + \sum_{l=r+1}^N C_l e^{s_l t}.$$

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How to determine the coefficients $C_1, C_2, \dots, C_N$?

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Need N auxiliary conditions. e.g.,

$$y(t), \frac{d}{dt}y(t), \dots, \frac{d^{N-1}}{dt^{N-1}}y(t) \quad \text{at} \quad t = t_0$$

Method of undetermined coefficients (5)

- Different choices for these auxiliary conditions result in different input-output relationships. (Textbook, Problem 2.34)
- Depending on how the auxiliary information is stated or whether the auxiliary information is available, the system may or may not correspond to a linear system, may or may not correspond to a linear time-invariant (LTI) system, may or may not correspond to a causal and LTI system.
- We focus on differential equations used to describe systems that are LTI and causal.

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Initial rest states that if $x(t) = 0$ for $t \leq t_0$, then $y(t) = 0$ for $t \leq t_0$.

In words the output must be zero up until the time when the input becomes nonzero.

Causal and LTI system $\iff$ initial rest

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Initial rest (1)

- It is relatively straightforward to see

Causal and LTI system $\implies$ initial rest

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intuition as to why initial rest $\implies$ time-invariant system.

If we perform **identical experiments** on two successive days, where the **circuit starts from initial rest** at noon on each day, then we would expect to see **identical responses**, *i.e.*, responses that are simply **time-shifted** by one day with respect to each other.

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initial/auxiliary conditions = 0

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- The initial conditions at $t = t_0^-$ must be translated to $t = t_0^+$ to reflect the effect of applying the input at $t = t_0$.
- A necessary/sufficient condition: the right-hand side of the differential equation contains **no impulses or derivatives of impulses**.
- If not, the initial conditions at $t = t_0^-$ are no longer equal to the initial conditions at $t = t_0^+$ (**out of scope of this course**).

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$$y(t_0) = \frac{d}{dt}y(t_0) = \dots = \frac{d^{N-1}}{dt^{N-1}}y(t_0) = 0$$

- The initial conditions at $t = t_0^-$ must be translated to $t = t_0^+$ to reflect the effect of applying the input at $t = t_0$.
- A necessary/sufficient condition: the right-hand side of the differential equation contains **no impulses or derivatives of impulses**.
- If not, the initial conditions at $t = t_0^-$ are no longer equal to the initial conditions at $t = t_0^+$ (out of scope of this course).

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t)$$

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The method of undetermined coefficients

- The **method of undetermined coefficients** assumes that the **forced response** or **particular solution** is the sum of functions of the mathematical form of the excitation $x(t)$ and all (**distinct, nonzero**) derivatives of $x(t)$ that differ in form from $x(t)$ *i.e.*,

$$y_p(t) = P_0 x(t) + P_1 \frac{d}{dt} x(t) + \dots$$

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Forced response examples

Example

1

$$x(t) = 5e^{-7t}u(t) \implies y_p(t) = P_0e^{-7t}u(t)$$

2

$$x(t) = 170 \cos(377t) u(t) \implies$$
$$y_p(t) = [P_0 \cos(377t) + P_1 \sin(377t)] u(t)$$

3

$$x(t) = 10u(t) \implies y_p(t) = P_0u(t)$$

Method of undetermined coefficients (7)

Combining the **natural response** and **forced response**, the overall solution has the form

$$\begin{aligned} y(t) &= y_h(t) + y_p(t) \\ &= \sum_{l=1}^N C_l e^{s_l t} + P_0 x(t) + P_1 \frac{d}{dt} x(t) + \cdots \end{aligned}$$

which we plug into the differential equation and then solve for the **undetermined coefficients** (the C_l 's and the P_k 's).

Impulse response of LTI diffeq systems (1)

- In a diffeq class, the focus is on computing solutions to the diffeq for various input signals.
- In systems analysis, the diffeq indirectly describes the **input-output relationship** of an LTI system, and we are more interested in finding the **impulse response $h(t)$** of that system (from which we can then look at **stability** and other **system properties**).

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Impulse response of LTI diffeq systems (2)

- In principle we could find $h(t)$ by letting $x(t) = \delta(t)$ and solving the diffeq.
- Typically this time-domain approach requires more work and is less insightful than the frequency-domain / PFE approach described later.
- Nevertheless, for completeness we work one example here.
- In this example, we simplify things by first finding the step response, and then differentiating to find the impulse response, as in earlier discussion of step response.

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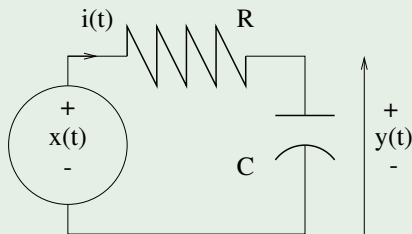
Example (1)

Example

Find step response and impulse response of the following RC circuit, which had diffeq

$$y(t) + \frac{1}{\alpha} \frac{d}{dt} y(t) = x(t)$$

for $\alpha = 1/RC$.



Solution (1)

Find **natural response** $y_h(t) = Ce^{st}$.

$$y_h(t) + \frac{1}{\alpha} \frac{d}{dt} y_h(t) = 0$$

$$\implies Ce^{st} + \frac{1}{\alpha} s Ce^{st} = 0$$

$$\implies s + \alpha = 0, \quad (C \neq 0, e^{st} \neq 0)$$

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Solution (2)

We analyze a unit-step input signal

$$x(t) = u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}.$$

Since $x(t) = 0$ for $t \leq 0$, the condition of **initial rest** implies the auxiliary condition **$y(t) = 0$ for $t \leq 0$** .

Since $x(t) = 1$, $\frac{d}{dt}x(t) = 0$ for $t \geq 0$, the forced response is

$$y_p(t) = P \text{ for } t \geq 0$$

Thus

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Solution (3)

Plugging into the diffeq yields:

$$y(t) + \frac{1}{\alpha} \frac{d}{dt} y(t) = x(t)$$

$$\Rightarrow [Ce^{-\alpha t} + P] + \frac{1}{\alpha} [-\alpha Ce^{-\alpha t}] = 1 \quad (\text{for } t \geq 0)$$

$$\Rightarrow P = 1.$$

Furthermore, using the initial condition

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Solution (4)

$$y_h(t) = -e^{-\alpha t} \text{ and } y_p(t) = 1 \text{ for } t \geq 0.$$

Note that

$$y_h(t) \rightarrow 0 \text{ as } t \rightarrow \infty, \quad y(t) \rightarrow y_p(t) \text{ as } t \rightarrow \infty,$$

These effects are called the **transient response** and the **steady-state response**

Our final solution for the step response is

$$y(t) = \begin{cases} [1 - e^{-\alpha t}], & t \geq 0 \\ 0, & t < 0 \end{cases} = [1 - e^{-\alpha t}]u(t).$$

so taking the derivative yields the impulse response

$$h(t) = \alpha e^{-\alpha t} u(t) = \frac{1}{RC} e^{-t/RC} u(t).$$

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- We have the impulse response, we can determine the response of the system to any other input signal simply by performing convolution, rather than re-solving the differential equation!
- Find $h(t)$ by time-domain methods was not too painful in this case, but it gets laborious for higher-order systems. The Fourier and Laplace methods greatly simplify.

Solution (5)

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Summary of diffeqs

- Main point is hardware implementation with RLC circuits yields systems described by diffeqs. Thus this class of systems is extremely important.
- Analog filter design is partly about how to make efficient approximations to a desired impulse response! (n and m limited by cost/complexity)
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Outline

1 2. CT LTI Systems

- Overview
- Introduction
- Techniques for the analysis of linear systems
- Impulse response: mathematical and physical introduction
- Impulse representation of CT signals (2.2.1)
- Convolution for CT LTI systems (2.2.2)
- Properties of convolution and LTI systems (2.3)
- LTI system properties via impulse response (2.3.4-7)
 - T-1 Causal LTI systems (2.3.6)
 - T-2 Memory (2.3.4)
 - A-2 Stability of LTI systems (2.3.7)
 - A-3 Invertibility of LTI systems (2.3.5)
- Step response (2.3.8)
- CT systems described by differential equation models (2.4)
 - Solution of linear constant-coefficient diffeqs (2.4.1)

● Summary

Summary

- impulse response
- convolution integral for LTI systems
- graphical convolution
- properties of convolution
- convolution and LTI system interconnection
- impulse response vs step response
- LTI system properties characterized by $h(t)$ (causality, memory, stability, invertibility)
- diffeq systems
- solutions of diffeqs